

EE114 Homework

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Problem 2.6.2

Given: From Problem 2.4.2, X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

So X is discrete with PMF: $P_X(-1) = 0.2$, $P_X(0) = 0.5$, $P_X(1) = 0.3$.

Let $V = g(X) = |X|$.

(a) **Find** $P_V(v)$.

Since $V = |X|$, the possible values of V are $\{0, 1\}$.

$$\begin{aligned} P_V(0) &= P[|X| = 0] = P_X(0) = 0.5, \\ P_V(1) &= P[|X| = 1] = P_X(-1) + P_X(1) = 0.2 + 0.3 = 0.5. \end{aligned}$$

Thus:

$$P_V(v) = \begin{cases} 0.5 & v = 0, \\ 0.5 & v = 1, \\ 0 & \text{otherwise.} \end{cases}$$

(b) **Find** $F_V(v)$.

$$F_V(v) = \begin{cases} 0 & v < 0, \\ 0.5 & 0 \leq v < 1, \\ 1 & v \geq 1. \end{cases}$$

(c) **Find** $E[V]$.

$$E[V] = 0 \cdot P_V(0) + 1 \cdot P_V(1) = 0(0.5) + 1(0.5) = \boxed{0.5}.$$

Problem 2.6.3

Given: The random variable X from Problem 2.4.3 (using the same X from 2.4.2 with $P_X(-1) = 0.2$, $P_X(0) = 0.5$, $P_X(1) = 0.3$). Let $W = g(X) = -X$.

(a) **Find** $P_W(w)$.

Since $W = -X$, the values of W are $\{-1, 0, 1\}$.

$$\begin{aligned}P_W(-1) &= P[X = 1] = 0.3, \\P_W(0) &= P[X = 0] = 0.5, \\P_W(1) &= P[X = -1] = 0.2.\end{aligned}$$

$$P_W(w) = \begin{cases} 0.3 & w = -1, \\ 0.5 & w = 0, \\ 0.2 & w = 1, \\ 0 & \text{otherwise.} \end{cases}$$

(b) **Find** $F_W(w)$.

$$F_W(w) = \begin{cases} 0 & w < -1, \\ 0.3 & -1 \leq w < 0, \\ 0.8 & 0 \leq w < 1, \\ 1 & w \geq 1. \end{cases}$$

(c) **Find** $E[W]$.

$$E[W] = (-1)(0.3) + (0)(0.5) + (1)(0.2) = -0.3 + 0 + 0.2 = \boxed{-0.1}.$$

Note: $E[W] = E[-X] = -E[X]$. Since $E[X] = (-1)(0.2) + 0(0.5) + 1(0.3) = 0.1$, we get $E[W] = -0.1$. ✓

Problem 3.7.1

Given: $X \sim \text{Uniform}(0, 1)$, so $f_X(x) = 1$ for $0 \leq x \leq 1$ and $F_X(x) = x$ for $0 \leq x \leq 1$. The instantaneous power across a 1Ω resistor is $Y = X^2$.

Find the CDF $F_Y(y)$ **and PDF** $f_Y(y)$.

For $0 \leq y \leq 1$:

$$F_Y(y) = P[Y \leq y] = P[X^2 \leq y] = P[X \leq \sqrt{y}] = F_X(\sqrt{y}) = \sqrt{y}.$$

(Since $X \geq 0$, $X^2 \leq y \iff X \leq \sqrt{y}$.)

So:

$$F_Y(y) = \begin{cases} 0 & y < 0, \\ \sqrt{y} & 0 \leq y \leq 1, \\ 1 & y > 1. \end{cases}$$

Differentiating:

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}} & 0 < y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Problem 3.7.2

Given: $X \sim \text{Exponential}(\lambda)$, so $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Let $Y = \sqrt{X}$.

Find the CDF and PDF of Y , and show Y is Rayleigh.

For $y \geq 0$:

$$F_Y(y) = P[Y \leq y] = P[\sqrt{X} \leq y] = P[X \leq y^2] = F_X(y^2) = 1 - e^{-\lambda y^2}.$$

$$F_Y(y) = \begin{cases} 0 & y < 0, \\ 1 - e^{-\lambda y^2} & y \geq 0. \end{cases}$$

Differentiating:

$$f_Y(y) = \begin{cases} 2\lambda y e^{-\lambda y^2} & y \geq 0, \\ 0 & y < 0. \end{cases}$$

The Rayleigh PDF with parameter a is:

$$f_Y(y) = \frac{y}{a^2} e^{-y^2/(2a^2)}, \quad y \geq 0.$$

Comparing: $2\lambda = \frac{1}{a^2}$ and $\lambda = \frac{1}{2a^2}$, so:

$$a = \frac{1}{\sqrt{2\lambda}}.$$

Problem 4.1.1

Given:

$$F_{X,Y}(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) $P[X \leq 2, Y \leq 3]$:

$$P[X \leq 2, Y \leq 3] = F_{X,Y}(2, 3) = (1 - e^{-2})(1 - e^{-3}) \approx (0.8647)(0.9502) \approx \boxed{0.8218}.$$

(b) **Marginal CDF $F_X(x)$:**

$$F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y) = (1 - e^{-x})(1) = \begin{cases} 1 - e^{-x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

(c) **Marginal CDF $F_Y(y)$:**

$$F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x, y) = (1)(1 - e^{-y}) = \begin{cases} 1 - e^{-y} & y \geq 0, \\ 0 & y < 0. \end{cases}$$

Problem 4.1.2

Express the following in terms of $F_X(x)$ and $F_Y(y)$.

- (a) $F_{X,Y}(x, -\infty) = 0$. (CDF evaluated as $y \rightarrow -\infty$ is always 0.)
- (b) $F_{X,Y}(x, \infty) = F_X(x)$. (Marginal CDF of X .)
- (c) $F_{X,Y}(-\infty, \infty) = 0$. (CDF as $x \rightarrow -\infty$ is 0.)
- (d) $F_{X,Y}(-\infty, y) = 0$. (CDF as $x \rightarrow -\infty$ is 0.)
- (e) $F_{X,Y}(\infty, y) = F_Y(y)$. (Marginal CDF of Y .)

Problem 4.2.2

Given:

$$P_{X,Y}(x, y) = \begin{cases} c|x+y| & x = -2, 0, 2; y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) **Find c .**

We compute $|x+y|$ for all (x, y) pairs:

$x \backslash y$	-1	0	1
-2	$ -3 = 3$	$ -2 = 2$	$ -1 = 1$
0	$ -1 = 1$	$ 0 = 0$	$ 1 = 1$
2	$ 1 = 1$	$ 2 = 2$	$ 3 = 3$

$$\text{Sum} = 3 + 2 + 1 + 1 + 0 + 1 + 1 + 2 + 3 = 14.$$

$$\text{So } 14c = 1 \implies \boxed{c = 1/14}.$$

- (b) $P[Y < X]$:

We need pairs where $y < x$:

- $x = -2$: no $y \in \{-1, 0, 1\}$ satisfies $y < -2$.
- $x = 0$: $y = -1$: $|0 + (-1)| = 1$.
- $x = 2$: $y = -1$: $|2 + (-1)| = \underline{1}$; $y = 0$: $|2 + 0| = \underline{2}$; $y = 1$: $|2 + 1| = \underline{3}$.

$$\text{Sum} = 0 + 1 + 1 + 2 + 3 = 7.$$

$$P[Y < X] = \frac{7}{14} = \boxed{\frac{1}{2}}.$$

- (c) $P[Y > X]$:

Pairs where $y > x$:

- $x = -2$: $y = -1$: 3; $y = 0$: 2; $y = 1$: 1. Sum = 6.
- $x = 0$: $y = 1$: 1. Sum = 1.
- $x = 2$: none.

$$\text{Sum} = 6 + 1 = 7.$$

$$P[Y > X] = \frac{7}{14} = \boxed{\frac{1}{2}}.$$

(d) $P[Y = X]$:

Pairs where $y = x$: only $(0, 0)$, where $|0 + 0| = 0$.

$$P[Y = X] = \frac{0}{14} = \boxed{0}.$$

(e) $P[X < 1]$:

$X < 1$ means $x \in \{-2, 0\}$.

For $x = -2$: $3 + 2 + 1 = 6$. For $x = 0$: $1 + 0 + 1 = 2$. Sum = 8.

$$P[X < 1] = \frac{8}{14} = \boxed{\frac{4}{7}}.$$

Problem 4.2.3

Given: Two integrated circuits are tested. The probability of rejecting each is p . X = number of rejects in test 1, Y = number of rejects in test 2. Both X and Y are $\in \{0, 1\}$.

Since the two tests are independent, each is a Bernoulli(p) trial. The joint PMF is:

$$P_{X,Y}(x, y) = \begin{cases} (1-p)^2 & (x, y) = (0, 0), \\ p(1-p) & (x, y) = (1, 0), \\ p(1-p) & (x, y) = (0, 1), \\ p^2 & (x, y) = (1, 1), \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, $P_{X,Y}(x, y) = p^{x+y}(1-p)^{2-x-y}$ for $x, y \in \{0, 1\}$.

Problem 4.3.1

Given: X and Y from Problem 4.2.1 with joint PMF $P_{X,Y}(x, y) = cxy$ for $x = 1, 2, 4$; $y = 1, 3$.

First, find c : compute $\sum_{x,y} cxy$:

$$\sum = c[1 \cdot 1 + 1 \cdot 3 + 2 \cdot 1 + 2 \cdot 3 + 4 \cdot 1 + 4 \cdot 3] = c[1 + 3 + 2 + 6 + 4 + 12] = 28c = 1.$$

So $c = 1/28$.

(a) **Marginal PMFs $P_X(x)$ and $P_Y(y)$:**

$$P_X(x) = \sum_y P_{X,Y}(x, y) = cx(1 + 3) = \frac{4x}{28} = \frac{x}{7}.$$

So $P_X(1) = 1/7$, $P_X(2) = 2/7$, $P_X(4) = 4/7$.

$$P_Y(y) = \sum_x P_{X,Y}(x, y) = cy(1 + 2 + 4) = \frac{7y}{28} = \frac{y}{4}.$$

So $P_Y(1) = 1/4$, $P_Y(3) = 3/4$.

(b) $E[X]$ and $E[Y]$:

$$E[X] = 1 \cdot \frac{1}{7} + 2 \cdot \frac{2}{7} + 4 \cdot \frac{4}{7} = \frac{1+4+16}{7} = \boxed{\frac{21}{7} = 3}.$$

$$E[Y] = 1 \cdot \frac{1}{4} + 3 \cdot \frac{3}{4} = \frac{1+9}{4} = \boxed{\frac{10}{4} = \frac{5}{2}}.$$

(c) **Standard deviations** σ_X and σ_Y :

$$E[X^2] = 1 \cdot \frac{1}{7} + 4 \cdot \frac{2}{7} + 16 \cdot \frac{4}{7} = \frac{1+8+64}{7} = \frac{73}{7}.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{73}{7} - 9 = \frac{73-63}{7} = \frac{10}{7}.$$

$$\boxed{\sigma_X = \sqrt{10/7} \approx 1.195.}$$

$$E[Y^2] = 1 \cdot \frac{1}{4} + 9 \cdot \frac{3}{4} = \frac{1+27}{4} = 7.$$

$$\text{Var}(Y) = 7 - \left(\frac{5}{2}\right)^2 = 7 - \frac{25}{4} = \frac{3}{4}.$$

$$\boxed{\sigma_Y = \sqrt{3/4} = \frac{\sqrt{3}}{2} \approx 0.866.}$$

Problem 4.3.2

Given: X and Y from Problem 4.2.2 with $P_{X,Y}(x,y) = \frac{1}{14}|x+y|$ for $x \in \{-2, 0, 2\}$, $y \in \{-1, 0, 1\}$.

Recall the joint PMF table (each entry multiplied by 1/14):

$x \backslash y$	-1	0	1	$P_X(x)$
-2	3/14	2/14	1/14	6/14 = 3/7
0	1/14	0	1/14	2/14 = 1/7
2	1/14	2/14	3/14	6/14 = 3/7
$P_Y(y)$	5/14	4/14 = 2/7	5/14	

(a) **Marginal PMFs:**

$$P_X(-2) = \frac{3}{7}, \quad P_X(0) = \frac{1}{7}, \quad P_X(2) = \frac{3}{7}.$$

$$P_Y(-1) = \frac{5}{14}, \quad P_Y(0) = \frac{2}{7}, \quad P_Y(1) = \frac{5}{14}.$$

(b) $E[X]$ and $E[Y]$:

$$E[X] = (-2) \cdot \frac{3}{7} + 0 \cdot \frac{1}{7} + 2 \cdot \frac{3}{7} = \frac{-6+0+6}{7} = \boxed{0}.$$

$$E[Y] = (-1) \cdot \frac{5}{14} + 0 \cdot \frac{2}{7} + 1 \cdot \frac{5}{14} = \frac{-5+0+5}{14} = \boxed{0}.$$

(c) **Standard deviations:**

$$E[X^2] = 4 \cdot \frac{3}{7} + 0 + 4 \cdot \frac{3}{7} = \frac{24}{7}. \quad \text{Var}(X) = \frac{24}{7} - 0 = \frac{24}{7}.$$

$$\sigma_X = \sqrt{24/7} = \frac{2\sqrt{6}}{\sqrt{7}} \approx 1.852.$$

$$E[Y^2] = 1 \cdot \frac{5}{14} + 0 + 1 \cdot \frac{5}{14} = \frac{10}{14} = \frac{5}{7}. \quad \text{Var}(Y) = \frac{5}{7}.$$

$$\sigma_Y = \sqrt{5/7} \approx 0.845.$$

Problem 4.4.1

Given:

$$f_{X,Y}(x,y) = \begin{cases} c & x+y \leq 1, x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) **Find c .**

The region is a triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$. Area = $1/2$.

$$\int_0^1 \int_0^{1-x} c \, dy \, dx = c \cdot \frac{1}{2} = 1 \implies \boxed{c = 2}.$$

(b) $P[X \leq Y]$:

The region where $x \leq y$ within the triangle:

$$P[X \leq Y] = \int_0^{1/2} \int_x^{1-x} 2 \, dy \, dx = 2 \int_0^{1/2} (1-2x) \, dx = 2 \left[x - x^2 \right]_0^{1/2} = 2 \left(\frac{1}{2} - \frac{1}{4} \right) = \boxed{\frac{1}{2}}.$$

(c) $P[X + Y \leq 1/2]$:

This is the triangle with vertices $(0,0)$, $(1/2,0)$, $(0,1/2)$. Area = $1/8$.

$$P[X + Y \leq 1/2] = 2 \cdot \frac{1}{8} = \boxed{\frac{1}{4}}.$$

Problem 4.4.2

Given:

$$f_{X,Y}(x,y) = \begin{cases} cxy^2 & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) **Find c .**

$$\int_0^1 \int_0^1 cxy^2 \, dy \, dx = c \int_0^1 x \, dx \int_0^1 y^2 \, dy = c \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{c}{6} = 1 \implies \boxed{c = 6}.$$

(b) $P[X > Y]$ **and** $P[Y < X^2]$:

$$P[X > Y] = 6 \int_0^1 \int_0^x xy^2 dy dx = 6 \int_0^1 x \cdot \frac{x^3}{3} dx = 2 \int_0^1 x^4 dx = 2 \cdot \frac{1}{5} = \boxed{\frac{2}{5}}.$$

$$P[Y < X^2] = 6 \int_0^1 \int_0^{x^2} xy^2 dy dx = 6 \int_0^1 x \cdot \frac{x^6}{3} dx = 2 \int_0^1 x^7 dx = 2 \cdot \frac{1}{8} = \boxed{\frac{1}{4}}.$$

(c) $P[\min(X, Y) \leq 1/2]$:

Use complement: $P[\min(X, Y) \leq 1/2] = 1 - P[\min(X, Y) > 1/2] = 1 - P[X > 1/2, Y > 1/2]$.

$$\begin{aligned} P[X > 1/2, Y > 1/2] &= 6 \int_{1/2}^1 \int_{1/2}^1 xy^2 dy dx = 6 \int_{1/2}^1 x dx \int_{1/2}^1 y^2 dy \\ &= 6 \cdot \frac{3/8}{1} \cdot \frac{7/24}{1} = 6 \cdot \frac{3}{8} \cdot \frac{7}{24} = 6 \cdot \frac{21}{192} = \frac{126}{192} = \frac{21}{32}. \end{aligned}$$

$$P[\min(X, Y) \leq 1/2] = 1 - \frac{21}{32} = \boxed{\frac{11}{32}}.$$

(d) $P[\max(X, Y) \leq 3/4]$:

$$\begin{aligned} P[\max(X, Y) \leq 3/4] &= P[X \leq 3/4, Y \leq 3/4] = 6 \int_0^{3/4} \int_0^{3/4} xy^2 dy dx \\ &= 6 \int_0^{3/4} x dx \int_0^{3/4} y^2 dy = 6 \cdot \frac{9}{32} \cdot \frac{27}{192} = 6 \cdot \frac{9}{32} \cdot \frac{9}{64} = 6 \cdot \frac{81}{2048} = \frac{486}{2048} = \boxed{\frac{243}{1024}}. \end{aligned}$$

Problem 4.5.2

Given:

$$f_{X,Y}(x, y) = \begin{cases} 2 & x + y \leq 1, x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) **Marginal PDF** $f_X(x)$:

For $0 \leq x \leq 1$:

$$f_X(x) = \int_0^{1-x} 2 dy = 2(1-x).$$

$$f_X(x) = \begin{cases} 2(1-x) & 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(b) **Marginal PDF** $f_Y(y)$:

By symmetry (same structure), for $0 \leq y \leq 1$:

$$f_Y(y) = \int_0^{1-y} 2 dx = 2(1-y).$$

$$f_Y(y) = \begin{cases} 2(1-y) & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Problem 4.5.3

Given: Over the circle $x^2 + y^2 \leq r^2$:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi r^2} & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) **Marginal PDF $f_X(x)$:**

For $-r \leq x \leq r$, y ranges from $-\sqrt{r^2 - x^2}$ to $\sqrt{r^2 - x^2}$:

$$f_X(x) = \int_{-\sqrt{r^2 - x^2}}^{\sqrt{r^2 - x^2}} \frac{1}{\pi r^2} dy = \frac{2\sqrt{r^2 - x^2}}{\pi r^2}.$$

$$f_X(x) = \begin{cases} \frac{2\sqrt{r^2 - x^2}}{\pi r^2} & -r \leq x \leq r, \\ 0 & \text{otherwise.} \end{cases}$$

(b) **Marginal PDF $f_Y(y)$:**

By symmetry:

$$f_Y(y) = \begin{cases} \frac{2\sqrt{r^2 - y^2}}{\pi r^2} & -r \leq y \leq r, \\ 0 & \text{otherwise.} \end{cases}$$